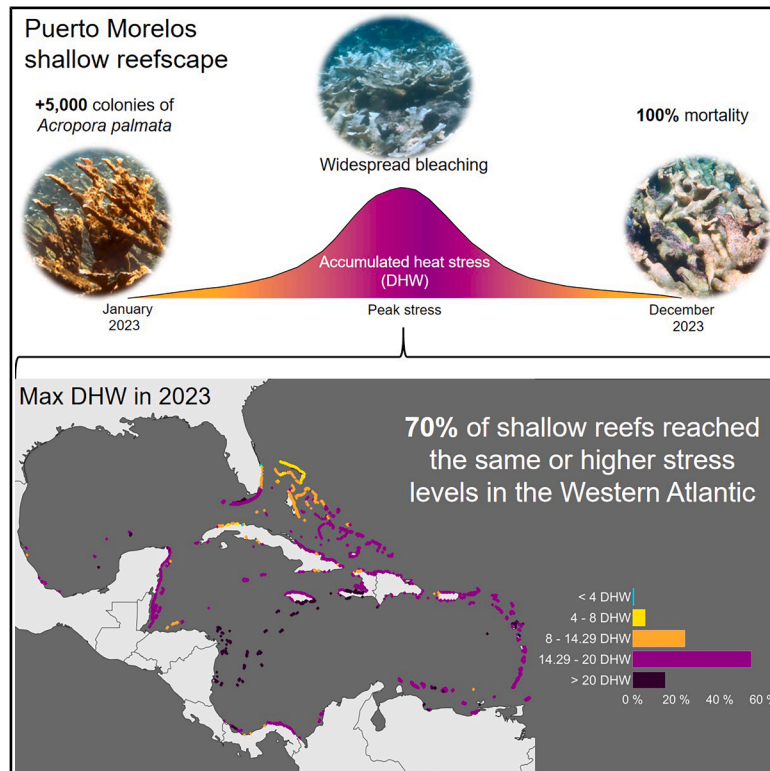


2023 global heatwave causes mass mortality of a keystone coral on shallow Western Atlantic reefs

Graphical abstract



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In brief

Environmental science; Environmental health; Environmental monitoring

Highlights

- Record-high sea surface temperatures in 2023 caused mass coral mortality
- The crucial reef-builder *Acropora palmata* vanished at the Puerto Morelos reefscape
- The majority of Western Atlantic reefs reached higher thermal stress than our site



Article

2023 global heatwave causes mass mortality of a keystone coral on shallow Western Atlantic reefs

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SUMMARY

In 2023, the world's oceans experienced record-high temperatures, particularly devastating Caribbean reefs. We use drone-derived imagery from before, during, and after the heatwave to quantify the spatial extent of bleaching and mortality of the primary reef-builder, *Acropora palmata*, across 170,000 m² of shallow reef in Puerto Morelos. We then estimate Caribbean-wide mortality trends by identifying the proportion of reef crests where thermal stress levels surpassed those observed at our site. In Puerto Morelos, 14.29° C-weeks resulted in 100% *A. palmata* mortality. Nearly 70% of reef crests in the Greater Caribbean reached equal or higher levels, suggesting a catastrophic mass mortality event of the species along its entire shallow-water distributional range. We encourage scientists and managers to assess damages to *A. palmata* populations and identify potentially heat-resistant genotypes for conservation and restoration. Our findings emphasize the need to abruptly reduce greenhouse gas emissions for coral reefs to have a future.

INTRODUCTION

The year 2023 was the warmest since global atmospheric temperature records began in 1880. Anomalously high temperatures caused an extreme marine heatwave associated with the long-term warming trend related to increasing greenhouse gas emissions and the 2023–2024 EL Niño event.^{1,2} The unprecedented heat stress prompted the longest and strongest global coral bleaching event in human history.^{3,4} Coral bleaching is a stress response due to the physiological disruption of the relationship between the coral host and their symbiotic algae which provide a significant part of the coral energy requirements. It can occur during events of prolonged levels of moderate stress, or as a result of acute stress if temperature intensity surpasses the thermal tolerance threshold of a coral.^{5,6} Between 2023 and 2024, nearly 75% of coral reefs worldwide were subjected to bleaching-level heat stress.⁷ The Greater Caribbean was particularly affected³ and experienced such high accumulative temperature anomalies that the Coral Reef Watch Program of the United States National Oceanic and Atmospheric Administration (NOAA) had to extend their categorical system for satellite-based bleaching alerts. Three new categories were added to consider the risk of severe and near complete multi-species mortality, with most Caribbean reefs reaching said alert levels between January of 2023 and April of 2024.^{8,9}

Among the negative consequences that this bleaching event brought upon all coral species, the case of the once main Caribbean reef-builder, *Acropora palmata*, is particularly worrisome.^{3,10} The species thrives in shallow waters where it forms large patches of great complexity.^{11,12} As such, it is the pillar of many ecosystem services including coastal protection, livelihoods, and more.^{13,14} *A. palmata* was heavily affected in the late 1970s due to the outbreak of white-band disease, with an estimated loss of 80%–98%.^{15,16} The species has recovered little since then, evoking great spatial heterogeneity of associated populations. Before the heatwave, some larger patches could still be found on reef crests and very scarcely distributed individual colonies on deeper reefs.^{17,18} However, the extreme temperatures of 2023–2024 have likely pushed *A. palmata* populations closer to collapse. A high degree of mortality has already been confirmed by reports from Florida, with a 77% loss of genotypic diversity¹⁹ and local extinction.^{20,21} The evidence shows mortality rates closely associated to thermal stress levels with a complete loss of all recorded colonies in the Western Keys and moderate colony mortality in the Northern section which experienced lower heat stress.^{19,22}

Despite the increasing concern regarding the ability of coral populations to withstand future levels of heat stress and sustain reef growth, we still lack crucial understanding of how severe stress levels operate at spatially exhaustive scales. This is



particularly relevant as most services associated with coral reefs, such as habitat provision and coastal protection, are related to the large-scale architectural complexity constructed over hundreds to thousands of meters,^{23,24} and not the fate of individual colonies. In this study, we evaluate the spatial extent of *A. palmata* mortality during the extreme heatwave of 2023–2024 across the shallow-water reefscape scale of the Greater Caribbean. We focus on reef crests (<5 m) as this is the primary habitat for *A. palmata*. First, drone imagery is analyzed of 170,000 m² of a shallow reef off the coast of Puerto Morelos that reached just over 14°C-weeks during 2023. We then extend our analysis to cover the complete shallow-water distributional range of *A. palmata* along the Caribbean Sea and Gulf of Mexico. For this, we extract spatial data of the distribution of reef crests from the Allen Coral Atlas²⁵ and superimpose the maximum degree heating week²² that each area reached in 2023 and 2024. Our aim is to estimate the possible spatial extent of *A. palmata* mortality as a result of the 2023 heatwave. Study outcomes are intended to serve local and regional monitoring efforts and aid in the identification of potentially heat-resistant populations or climate change refuges, with implications for the conservation and restoration of a species that may soon go extinct.

RESULTS AND DISCUSSION

Unusually high sea surface temperatures (SST) in Puerto Morelos began in May of 2023. The 24-h average quickly surpassed 30°C, as measured *in situ*,²⁶ with SST anomalies of 3°C, as compared to the historic mean of the satellite-derived NOAA climatology. By August, *in situ* SST reached its peak of nearly 35°C (Figure 1) and a temperature anomaly of 4.8°C. These values are unprecedented in the area, making the marine heatwave the worst ever recorded in Puerto Morelos.²² Acute heat stress in combination with prolonged temperature anomalies had catastrophic consequences for shallow-water coral communities. *A. palmata* populations vanished at the Puerto Morelos reefscape. Before the heatwave, we estimated with high overall accuracy (OA = 94%, κ = 0.9; Table S1) an area of 1,193 m² (1.3% of the reef matrix) that was covered by *A. palmata*. This translates to approximately 5,000 colonies within our orthomosaic and agrees with previous estimations of the population size of *A. palmata* on this reef.²⁷ In August of 2023, 57% of the species had already died and 43% was bleached (OA = 90%, κ = 0.88; Table S1). By July of 2024, no *A. palmata* presence was detected with 100% confidence (Table S1) which was confirmed by a systematic field validation conducted in September of 2024. We assessed a total of 38 colonies at 10 randomly computed points along the entire orthomosaic, all of which were covered exclusively by turf and red calcifying algae (Table S2). This indicates that mortality occurred several months before the survey. No alive tissue was found during a detailed evaluation of each colony. In fact, it is likely that the intensity of the heatwave caused such an acute stress response that rather than bleaching, many colonies simply “shed” their tissue.²⁸ This may also explain that by August of 2023, which was the temperature peak, over 50% of *A. palmata* tissue had already died (Figure 1).

Even though the genus *Acropora* has generally been considered thermally sensitive,^{6,29} *A. palmata* has shown surprising thermal resistance under temperature stress.^{28,30,31} Unfortunately, its resistance potential was insufficient during the extreme marine heatwave of 2023. The Western Florida Keys experienced a maximum accumulated heat stress of 14°C-weeks (some areas even higher)²² which led to 100% mortality of various genotypes.²⁸ The *A. palmata* population at Dry Tortugas, West Florida Shelf (14.35°C-weeks), was thought of as particularly resistant and completely vanished during a similar time period to the population at Puerto Morelos.²¹ Even though we do not have genomic data at Puerto Morelos, it is likely that our 170,000 m² orthomosaic that contained an estimation of over 5,000 colonies also included various genotypes,^{32,33} all of which died as a result of 14°C-weeks. Furthermore, Puerto Morelos exhibits an annual temperature variability of approximately 5°C^{34,35} which is relatively high compared to other areas in the tropical Atlantic.^{36,37} Generally, reefs that exhibit high annual SST variabilities are thought to contain more heat-resistant coral populations due to intergenerational acclimatization. This is because certain environmental conditions lead to genotype-independent methylation of genes that regulate stress in coral which, in turn, increases tolerance levels in offspring.^{38,39} These epigenetic mechanisms may be related to the transcription of certain proteins that regulate heat stress response and favor survival.⁴⁰ Since, *A. palmata* is most commonly associated with *Symbiodinium fitti*,^{41,42} differences in thermal tolerance would primarily be attributed to the coral host or, potentially, intra-genet variations from fine-scale genetic variations of the holobiont.²⁸ On rare occasions, *A. palmata* can form a symbiotic relationship with *Durisdinium trenchii* which is known for its increased heat-tolerance, however, these are exceptions.⁴³ As such, it is logical to assume that region-wide 14°C-weeks surpass the general thermal tolerance threshold of *A. palmata* as evidenced by 100% mortality of likely heat-tolerant populations in Puerto Morelos and Florida.

Continuing, we estimated the proportion of reef crests that experienced equal or higher temperature stress across the entire distributional range of *A. palmata*. We found that in 2023, nearly 70% of crests in the Greater Caribbean were subjected to the same level of heat stress or higher than Puerto Morelos. An alarming 15% reached the previous maximum of the DHW scale, set at 20°C-weeks, or higher. Areas affected include coral reefs of Southern Florida, Honduras and Nicaragua, as well as Jamaica and the Southern reefs of Haiti. Another 24% of crests were exposed to an accumulated maximum heat stress between 8 and 14°C-weeks which is thought to induce reef-wide mortality with major effects on heat-sensitive species.⁸ Only 6% of crests experienced 4 to 8°C-weeks and less than 1% were exposed to less than 4°C-weeks (Figure 2). In 2024, the Greater Caribbean was affected by heat stress levels of similar severity with most eastern Caribbean reefs experiencing even higher heat stress than in the previous year. Nevertheless, at those sites, the observed DHW maxima of 2023 had already surpassed *A. palmata* stress tolerance levels (>14°C-weeks). Furthermore, we found a spatial correlation between the maximum accumulated heat stress in 2023 and 2024, at least for shallow-water reefs (Figure S1,

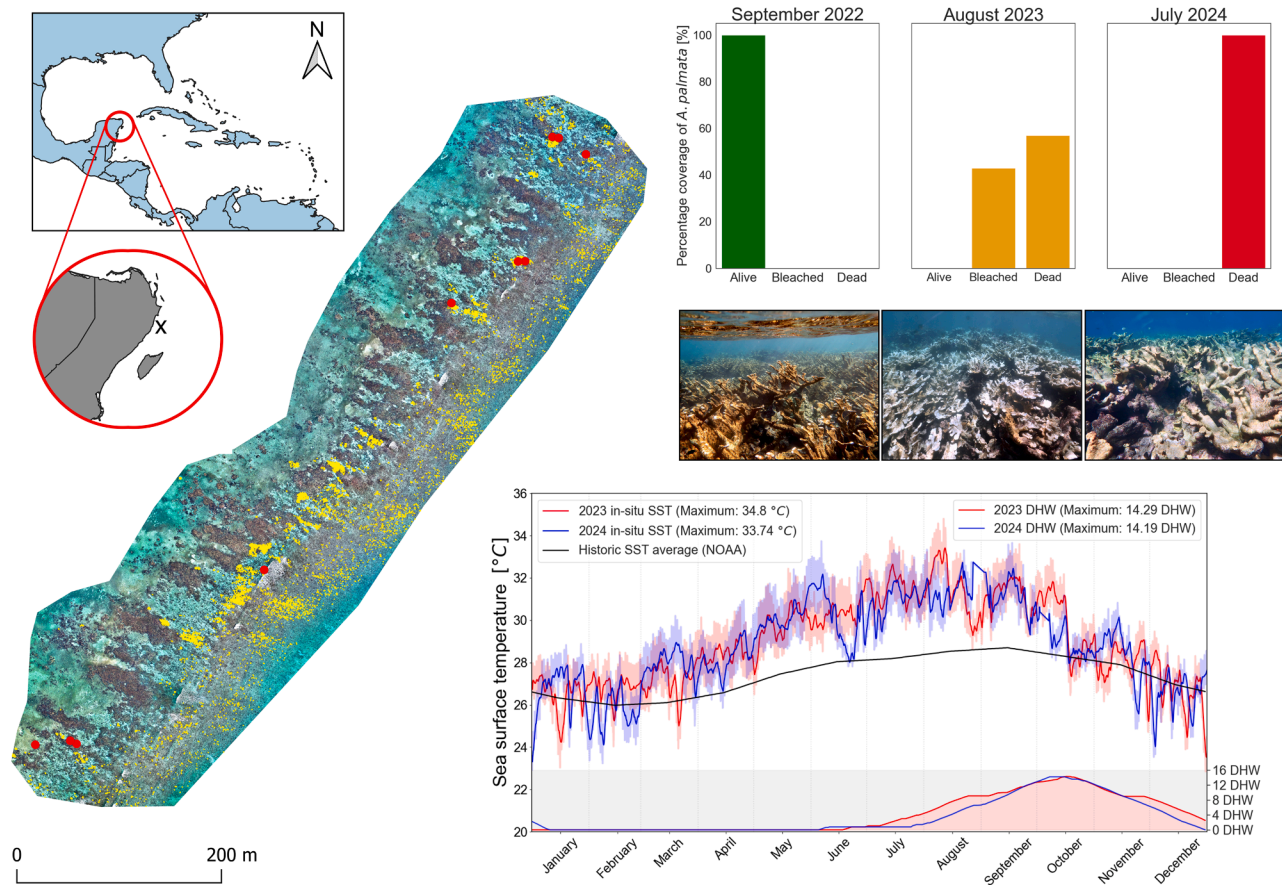


Figure 1. Thermal stress and *A. palmata* mass mortality at the Puerto Morelos reefscape during the 2023 marine heatwave

The shallow waters of Puerto Morelos (170,000 m²) contained 1,193 m² (=100%) of *A. palmata* before the heatwave (yellow in orthomosaic). In August of 2023, the area reached its maximum SST of 34.8°C (lower right image). Aerial data taken during this month indicated that 43% of *A. palmata* was bleached and 57% had already died, eventually increasing to 100% mortality by July of 2024. Classification results were verified in the field at ten randomly computed points that exhibited *A. palmata* presence in September of 2022 (indicated as red points in orthomosaic). Data to create the temperature curves was taken from SAMMO (for *in situ* data)²⁶ and NOAA Coral Reef Watch (for historic average and DHW values).²² *In situ* data show the 24-h SST average of Puerto Morelos in 2023 (red line) and 2024 (blue line), as well as hourly data (red/blue shadows) and associated degree heating week values.

$p < 0.01$, $r^2 = 0.64$). We, therefore, consider that the event of 2023 was the main driver of *A. palmata* mortality.

Virtually no *A. palmata* populations (<0.01%) were left untouched by the 2023 heatwave that exceeded heat tolerance thresholds in 2024, and those that survived are likely to exhibit genetic adaptations or have acclimatized to withstand such events.⁴⁴ In fact, we know of at least one shallow-water location, the iconic Limones reef which is located approximately 16 km Northeast of Puerto Morelos,⁴⁵ where satellite data indicates a heat stress of over 14°C-weeks, yet a major part of the sexually mature *A. palmata* population survived. This may be explained by two, not mutually exclusive, reasons. Firstly, we noted discrepancies between direct and remote SST data and associated anomalies in the Mexican Caribbean (Figure S2), a trend that has already been observed for this heatwave^{21,46} and previously in other locations.^{47–49} Secondly, *A. palmata* populations of Limones may exhibit increased resilience in the face of environmental stress.¹⁸ For example, associations with the endosymbiotic *D. trenchii* (clade D), even though uncommon in the

Caribbean, has shown a 1°C to 2°C increase in thermal tolerance of Floridian *A. palmata*.⁴³ This accords with previous studies that suggest that the endosymbiont, and particularly *D. trenchii*, may allow for the holobiont to survive under increased temperature stress as it has a much greater capacity to acclimatize.^{50,51} However, the exact reason behind why the majority of the *A. palmata* population at Limones reef was able to withstand the 2023 heatwave is still unknown. It is likely that more such reefs/populations exist for which, in the eye of the fourth and most severe global bleaching event, it is crucial to assess the population status of the critically endangered *A. palmata* through spatially explicit distributional and oceanographic data. We urge scientists and reef managers to assess associated damages and identify potentially heat-resistant genotypes and climate change refuges. This will allow for informed decision-making regarding the conservation and restoration of the species.

The likely widespread losses of *A. palmata* due to the 2023 heatwave throughout its entire shallow-water distributional range are catastrophic and the single most devastating event

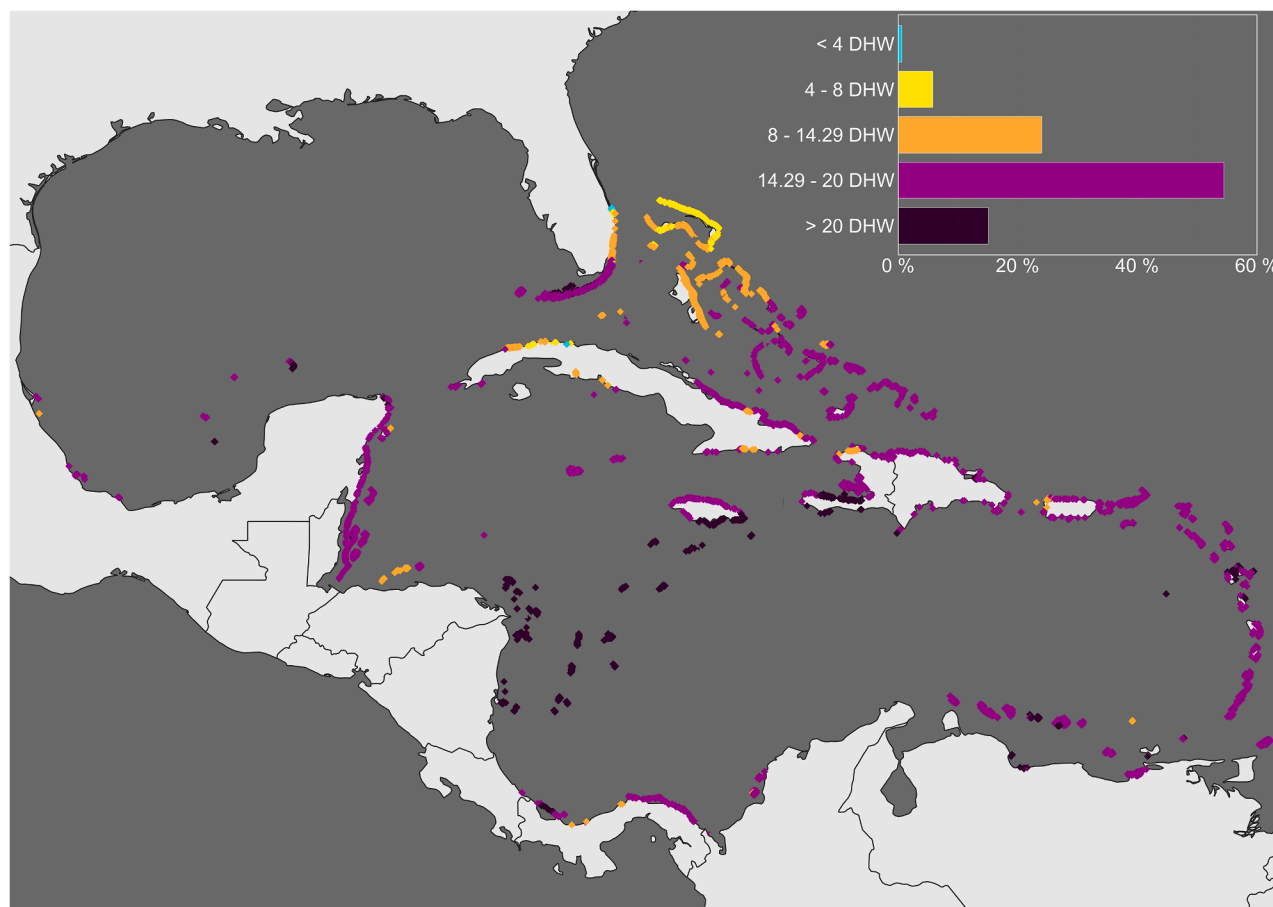


Figure 2. Maximum annual degree heating weeks at reef crests across the Caribbean Sea and Gulf of Mexico in 2023

Puerto Morelos reef and another 69.6% of shallow potential *A. palmata* habitat experienced a maximum of 14.29°C-weeks or higher (light/dark purple). A total of 24.0% and 5.8% reached 8 to 14.29 (orange) and 4 to 8°C-weeks (yellow), respectively. Data to create the map was taken from the Allen Coral Atlas²⁵ and NOAA's Coral Reef Watch 5 km annual maximum degree heating week (v3.1).²²

for the species since white band disease diminished its populations in the late 1970s. Until now, few reports confirm *A. palmata* survival, and in those areas the remaining colonies are separated by great distances.³ This implies virtually no fertilization success between them^{52,53} and a probable failure of natural population recovery. These are devastating news, since *A. palmata* was historically the main contributor toward reef-building processes, particularly on reef crests, and of utmost geological and ecological importance.^{10,54} Repercussions are vast, including reduced coastal protection capacity due to elevated water depths above coral reefs as reef-building processes (calcification) cannot keep up with sea level rise.²³ Furthermore, the structural complexity provided by *A. palmata* is known to sustain the biodiversity of key taxonomic groups that influence ecosystem processes which are crucial to reef functioning.^{55,56}

Climate change is the major cause behind species extinction,⁵⁷ giving way to an era of ecocide.^{58,59} Considering the trajectory of global SST anomalies, it comes as no surprise that the majority of reefs have experienced wide-spread mortality at least once in the last four decades.⁶⁰ At the same time, the spatial scale at which bleaching is occurring has increased drastically. The current inten-

sity and regularity of mass bleaching events, which in the Western Atlantic are two to three times more common than in other regions, do not allow for adequate ecosystem recovery after disturbance.⁶¹ As such, under the “business-as-usual” scenario, a coral reef mass extinction event is inevitable,³ evidence of the destructive effects that humans have had on these ecosystems. The future of our reefs will primarily depend on the trajectory of global carbon emissions⁶¹ and the ability of governments to hold themselves accountable for their destructive actions. For example, China, the United States, India, the European Union and Russia alone accounted for 64% of the world's CO₂ emissions in 2023.⁶² The inability, or rather unwillingness, to reduce our environmental footprint is mere negligence and borders on psychopathy, considering that the environmental integrity of coral reefs, and as we consider all ecosystems, is a human right.⁶³

Limitations of the study

We obtained aerial imagery of Puerto Morelos reef in 2022, 2023, and 2024. The extreme heat stress during summer of 2023 was the only major event that occurred during this time, however, it cannot fully be excluded that some colonies died due to other

reasons than the thermal stress. Nonetheless, we argue that those colonies died before the temperature peak of 2023. Additionally, it is difficult to make mortality predictions solely based on temperature data due to the possibility of individual colonies or potentially populations with temperature-resistant genetic traits (though probably isolated cases). Hence, there is hope that some colonies/populations survived heat stress the others could not. Finally, including more reefs into the analysis could improve the predictive power of our models. Further studies may address this shortcoming.

RESOURCE AVAILABILITY

Lead contact

Any additional information or requests will be addressed by the lead contact, Lara Virginia Birkart (lara.birkart@unam.edu).

Materials availability

This study did not generate new material.

Data and code availability

- Principal data have been shared in the manuscript. In the supplementary material we provide vector files of the classification results of each year as shape files (with explication), confusion matrices reporting overall classification accuracies of each year, field data taken during data verification, spatial correlation between the 2023 and 2024 maximum degree heating week values, discrepancies between satellite-derived and *in situ* SST data and associated anomalies, and a workflow of the methodological approach. Temperature data used in this study is freely available from the sources cited.
- The study does not report any original code.

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AUTHOR CONTRIBUTIONS

L.V.B. and L.A.-F. designed the study and collected data; L.V.B. analyzed the data; L.V.B. and L.A.-F. wrote and agreed upon the final version of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- METHOD DETAILS
 - Image acquisition and analysis
 - Estimating the extent of *A. palmata* mortality
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Raw and analyzed data	This paper	ISCIENCE-D-25-06898R2
Image classifications	This paper, supplementary zip file	ISCIENCE-D-25-06898R2
<i>in situ</i> temperature data, Puerto Morelos	SAMMO	https://sammo.icmyl.unam.mx/sammo.php
Satellite-derived temperature data and heat stress	NOAA Coral Reef Watch	https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW.html
Software and algorithms		
QGIS Desktop 3.28.2-Firenze	QGIS	https://qgis.org
PyCharm Community Edition 2023.1	Jetbrains	https://www.jetbrains.com/pycharm/
Agisoft Metashape Professional Version 1.8.3	Agisoft LLC	Node-Locked Educational License, 14331
Other		
Autel Evo II Dual 640t V2	Autel Robotics Co.	https://www.autelrobotics.com/productdetail/evo-2-dual-640t-v2-drones/
DJI Mavic 3 Enterprise	DJI	https://enterprise.dji.com/es/mavic-3-enterprise
GPSMAP 64s	Garmin	https://www.garmin.com/en-US/p/140022/

METHOD DETAILS

Our primary analyses focused on approximately one kilometer of crest along the Puerto Morelos reef as we monitored the area through high-resolution drone-derived aerial imagery since September of 2022. It forms part of the shallow reef of the Puerto Morelos Marine National Park which is characterized by its oligotrophic conditions. The park represents a critical habitat for the species *A. palmata* which, historically, dominated the shallow reefscape (<5 m).⁶⁴

Image acquisition and analysis

We obtained aerial imagery in September of 2022 (before heat stress), August of 2023 (temperature peak) and July of 2024 (after heat stress), covering approximately 170,000 m² of Puerto Morelos reef three times with a commercial drone. Flight missions were planned at a height of 60 m with an image overlay of 80% during the early morning hours to avoid sun glint and days with little to no wind and cloud coverage. We then created one rectified orthomosaic for each year with a spatial resolution of 2.0 cm per pixel using the photogrammetric software Agisoft Metashape Professional (v. 1.8.3), giving a total of three orthomosaics. Each was exported into QGIS (v. 3.28) in TIFF format in order to analyze the two-dimensional loss of *A. palmata* at the Puerto Morelos reefscape that occurred during the period before (September of 2022) and after (July of 2024) the heatwave.

The benthic substrate of each orthomosaic was classified through Object-Based Image Analysis (OBIA) into six classes of interest, including (1) live *A. palmata*, (2) bleached *A. palmata*, (3) dead *A. palmata*, (4) other bleached substrates, (5) other not-bleached substrates and (6) seagrass/sand. Following the methodologies proposed by Birkart et al. (2025), we used the Orfeo Toolbox in QGIS to segment the 2023 orthomosaic as it contained the most number of classes, applying the *meanshift* algorithm. Afterward, we trained a linear kernel type support vector machine (LIBSVM) classifier, giving 100 training points per class to classify the entire orthomosaic. The resulting shape file contained 688,406 objects, each with its corresponding class that reflects the underlying substrate (example given in Figure S3). Following, we manually re-classified those segments that changed their class dependencies throughout the three years. In specific, all segments associated with bleached *A. palmata*, dead *A. palmata* or other bleached substrate were re-categorized in accordance with the 2022 and 2024 orthomosaic. The new classes were live *A. palmata* in the case of 2022, dead *A. palmata* in the case of 2024, and bleached substrate changed to other not-bleached substrate. Finally, we created categorical maps of the three consecutive years and computed the area (m²) of each class in each orthomosaic, as well as associated changes.

Classification results were compared against two validation methodologies, firstly, the confusion matrix (Table S1) and, secondly, a field validation (Table S2). The confusion matrix is commonly used to verify the overall accuracy of (semi-)automatic classification processes. Here, in each of the three orthomosaics we randomly computed 50 points within each present class and compared the classification outcome to a by-eye interpretation of the underlying substrate. A kappa index of 0.81 or higher, as suggested by Landis and Koch,⁶⁵ which is calculated from the confusion matrix was considered as a successful result. Due to the easily identified footprint of live *A. palmata* and bleached organisms in high resolution RGB imagery, a visual inspection was sufficient to obtain

the classes for the training and validation points. Once the confusion matrix confirmed a high degree of agreement, we identified all *A. palmata* patches larger than 1 m² in size that were alive in September of 2022 and randomly computed ten points within those patches using the QGIS Python console for a field validation in September of 2024. The aim was to verify if colonies had died completely or little patches, not visible in the orthomosaic, had survived. We considered this as a necessary step to conclusively confirm the classification results of 100% *A. palmata* mortality because the spatial resolution of large-area imagery often does not allow for the identification of smaller objects, in this case potential partial survival of *A. palmata* tissue. Furthermore, alive tissue may have been present underneath colony branches which would be impossible to detect in orthomosaics due to the two-dimensional aerial view that drone images are taken at. GPS locations were extracted and considered the center of ten 10 m² circles in each of which we recorded the number of *A. palmata* colonies and the percentage of live tissue. Circles did not exhibit spatial overlap. We found a total of 38 colonies within the 100 m² assessed (Table S2). Finally, we estimated the number of colonies that died due to the heatwave in our orthomosaic. For this, we used historic data taken during 2014 where a total of 527 *A. palmata* colonies were measured in the field in Puerto Morelos, recording their length, width, height and health status. The area of each colony was calculated based on the formula of an ellipse, using its length and width. The total area of live *A. palmata* we obtained in September of 2022 (1,193 m²) was then divided by the average colony size (mean = 0.238 m², SD = 0.314 m²) to estimate the approximate number of colonies that were present before the heatwave.

Estimating the extent of *A. palmata* mortality

In order to estimate the spatial extent of *A. palmata* mortality across its potential habitat range, we obtained data of the distribution of reef habitat from the Allen Coral Atlas (2022). We limited our analysis to shallow waters of the Caribbean Sea and Gulf of Mexico, including solely the geomorphic zone *reef crest*, defined as generally shallow and characterized by highest wave energy absorbance,²⁵ for several reasons. Firstly, *A. palmata* exhibits highest population densities in this reef zone, forming nearly mono-specific patches of large spatial extents. Rarely, individual colonies can be found in deeper waters. Secondly, we took and analyzed aerial data at the shallow-water reefscape. Associated mortality trends may not be adequate for deeper waters as they can serve as thermal refuges. Following, we obtained the maximum annual DHW values of 2023 and 2024 from NOAA Coral Reef Watch.²² This product indicates the maximum accumulated heat stress that each 5 km by 5 km grid unit experienced in a given year. We superimposed both spatial files and identified the maximum DHW that all reef crests of the Greater Caribbean experienced in 2023 and 2024. Finally, we categorized reefs depending on the stress they experienced by adopting the categorical system of NOAA Coral Reef Watch⁸ and adjusting it to our observed *A. palmata* thermal threshold of 14°C-weeks, assuming comparable mortality to that observed at Puerto Morelos (Figure S3).

QUANTIFICATION AND STATISTICAL ANALYSIS

Geospatial analyses were performed in QGIS version 3.28.2-Firenze, in particular the Orfeo Toolbox (OTB) plugin for image classification. Statistical models were created in PyCharm Community Edition 2023.1 using the Python libraries *pandas* and *scipy*, as well as *matplotlib* and *seaborn* for data visualization. Generalized linear models are presented with their respective *p*-values, coefficient of determination (*r*²) and 95% confidence interval.